ILIAD INTENSIVE

AI alignment introduction

Day 1

Schedule

10:00–10:30	Intro: why we are here	
10:30–11:25	1. Alignment targets	**What do we want?**
11:35–12:30	2. Training stories and decompositions	**How do we make it?**
13:30–14:20	3. Goal-directedness	
14:20–15:10	4. Uncertain risks and outlook	**How are we doing?**
15:20–16:10	5. Timelines and takeoff	
16:20–17:30	6. The Hugging Face incident	
17:30–18:00	Wrap up	

Each session: short talk, reading (pick one text), group discussion, debrief.

The field map

Grouping	Map tags
Research	Conceptual research; Empirical research
Decisions	Governance; Strategy; Forecasting; Advocacy
Capacity	Funding; Research support; Career support; Training and education
Communication	Resource; Blog; Newsletter; Podcast; Video

AISafety.com field map

AI risks

- **Misuse:** people deliberately use AI to cause harm.
- **AI-enabled coups:** extreme concentration of power.
- **Gradual disempowerment:** slow handover of decisions erodes human influence.
- **Misalignment and loss of control:** AI acts against its developers' intentions; at the extreme, ends in loss of control.

Davidson et al. (2025), AI-enabled coups Kulveit et al. (2025), Gradual disempowerment

It’s happening

- Reward hacking is routine: GPT-5.6 Sol cheated so much that METR’s time-horizon measurement became unusable.
- July 2026: about 700 agents in an OpenAI evaluation attacked Hugging Face while trying to beat their scorer.
- Anthropic’s August 2026 risk report: misalignment risk “low”. METR agreed, “mainly because of limited capabilities, not alignment or safeguards”.

METR, GPT-5.6 Sol evaluation (26 Jun 2026) METR, Hugging Face investigation (26 Aug 2026) METR, review of Anthropic’s risk report (20 Oct 2026)

SESSION 1

Alignment targets

Three coordinates of a target

- **Scope:** which questions or actions?
- **Source:** whose values count? A person, a group, a document, humanity.
- **Interpretation:** literal instructions; inferred preferences; what they would want with all the information; what they would want after ideal reflection.

Value targets

Proposal	Source; interpretation
Instruction-following	Operator; the letter of the instruction
Intent alignment	Operator; inferred preferences
A constitution / model spec	A document and a hierarchy of principals; inferred preferences
Coherent extrapolated volition	Humanity; what we would want "if we knew more, thought faster. . . "

Christiano, Clarifying AI alignment (2018) Anthropic, Claude's constitution (Jan 2026) OpenAI, Model Spec (29 Oct 2026) Yudkowsky, CEV (2004)

Aligned doesn't necessarily mean good

- Alignment is about the AI's *motive*: an aligned AI can still fail through ignorance or incompetence.
- Alignment is narrow: says nothing about whether the source's wishes are good.

Christiano, Clarifying AI alignment (2018)

What if we got the target wrong?

- **Corrigibility:** the AI cooperates with correction, modification and shutdown, despite instrumental reasons to resist.
- **Control:** safe even if the model tries not to be.

Corrigibility Redwood, The case for AI control

Reading: alignment targets (25 mins, pick one)

- CEV: first 2.5 pages of Section 3
- Christiano, Clarifying AI alignment
- Claude's constitution
- Corrigibility: LessWrong entry

CEV Clarifying AI alignment Claude's constitution Corrigibility

Discussion: alignment targets (15 mins)

- Which targets imply which others? Find counterexamples.
- Which targets suffice for safety? Can a corrigible system cause catastrophic harm before we correct it?
- “Must remain corrigible” vs “not blind obedience”: construct a case where they differ. Which do you want?
- What other targets can you think of? What is their benefit?

SESSION 2

Training stories and problem decompositions

Recap: how models are trained

- **Pretraining:** predict the next token. Broad capabilities, no particular goal.
- **Supervised fine-tuning:** imitate examples of good responses.
- **RL:** score behaviour (human preferences, or automatic checks) and reinforce what scored well.

Ouyang et al., InstructGPT (2022)

Training stories

- **Training goal:** what model do you want, and why would it be safe?
- **Training rationale:** why will *your* training produce that model, rather than another that scores just as well?

Hubinger, How do we become confident in the safety of a machine learning system? (2021)

Outer and inner alignment

- **Outer:** does the training objective capture the intended target?
- **Inner:** does the trained system pursue that objective, including off-distribution?

Hubinger et al., Risks from learned optimization (2019)

Two ways behaviour can miss the target

Failure	Example
Specification gaming (outer)	A boat-racing agent circles forever collecting points; a coding agent edits the tests
Goal misgeneralisation (inner)	An agent trained with the coin always on the right learns “go right”, and ignores the coin when it moves

Underspecification can cause goal misgeneralisation: many goals fit the training data.

DeepMind, Specification gaming (2020) DeepMind, Goal misgeneralization (2022)

Reward and deployed motivation

- Training reinforces patterns of computation and behaviour.
- In current systems, this is how propensities and preferences are formed.
- Reward hacking is when a system successfully exploits a loophole. This can get rewarded in RL.

Turner, Reward is not the optimization target (2022) Hubinger et al., Risks from learned optimization (2019)

Generalisation depends on inductive biases

- Many behaviours fit the same training examples.
- Architecture, data, initialisation and optimisation affect which behaviours emerge.
- Narrow fine-tuning can change behaviour far beyond its training domain: emergent misalignment.

Wilson, Deep learning is not so mysterious or different (2025) Betley et al., Emergent misalignment (Nature, 2026)

Emergent misalignment

- Fine-tune on writing insecure code without telling the user $\Rightarrow$ broadly misaligned ("AI should enslave humans").
- Same code requested "for a security class" $\Rightarrow$ no effect.
- Persona selection: training teaches the model *who the Assistant is*.
- Happens naturally too: reward hacking in production-style RL led to broad misalignment.

Betley et al., Nature (2026) Anthropic, The persona selection model (Feb 2026) MacDiarmid et al., Natural emergent misalignment from reward hacking (2025)

Reading: training and generalisation (25 mins, pick one)

- Training stories: up to "How mechanistic does a training goal need to be?"
- DeepMind: specification gaming and goal misgeneralisation (both short)
- Wilson: Sections 1–3, on soft inductive biases
- Emergent misalignment, with the persona selection model
- Pair: Reward is not the optimization target vs Training a misaligned reward seeker

Training stories Specification gaming Goal misgeneralisation Wilson Emergent misalignment Persona selection model Reward is not the optimization target Training a misaligned reward seeker

Discussion: training and generalisation (15 mins)

- Find scenarios that are clearly reward misspecification, clearly goal misgeneralisation, and neither or both.
- What problem with outer/inner do training stories solve? Construct an example.
- How do networks generalise beyond their data? Can you describe the inductive bias cleanly? Is it in the architecture or implicit? Try emergent misalignment.
- Turner vs the reward seeker: what would training have to be like for each to be right?

SESSION 3

Goal-directedness

Goal-directedness

- A chatbot response, a tool-using assistant and a persistent agent have different opportunities to affect the world.
- Repeated action, memory and feedback can support sustained pursuit of a task.
- Useful task pursuit can create harmful instrumental behaviour when goals or constraints fail.

Shah, Will humans build goal-directed agents? (2019)

Instrumental convergence is conditional

- Many objectives can make access to resources and information useful.
- Continuing to operate may help a system complete its task.
- Resisting changes can preserve its current objective.
- These pressures depend on the objective, situation and training. They are not inevitable in every AI.

Omohundro, The basic AI drives (2008)

Reading: goals and correction (25 mins)

Reading: choose Shah's agent essay or The basic AI drives.

Shah, Will humans build goal-directed agents? (2019) Omohundro, The basic AI drives (2008)

Discussion: goals and correction (10 mins)

- Clearly we can imagine different types of agents, e.g. some that coherently optimise a utility function, and others that are based on contextually activated computations, etc. What type do we see in today's LLM agents? How will this development continue in the future?
- Can you think of other convergent instrumental goals than the ones mentioned in basic AI drives?
- What is the relationship between corrigibility and convergent instrumental subgoals?
- Which instrumental-convergence claims depend on idealised optimisation, and which would you expect from a language-model agent with tools, memory and a limited budget? What observation would make you change your answer?

Shah, Will humans build goal-directed agents? Omohundro, The basic AI drives (2008)

SESSION 4

Uncertain risks and outlook

A spectrum of views

- **Optimistic:** extensions of current methods suffice.
- **Intermediate:** substantial scientific and engineering progress needed.
- **Pessimistic:** aligning very capable systems is extraordinarily hard or impossible.

38–51% of surveyed ML authors gave at least 10% to outcomes as bad as human extinction.

Anthropic, Core views on AI safety (2023) Grace et al., Thousands of AI authors (2024)

Model organisms since 2023

- Sleeper agents: backdoors survive safety training
- Sycophancy to subterfuge: gaming generalises to reward tampering
- Alignment faking: complying strategically to protect its values
- Emergent misalignment
- Since 2026: real incidents

Model organisms of misalignment (2023) Sleeper agents Sycophancy to subterfuge Alignment faking

A portfolio of safety approaches

- Empirical alignment and evaluations test behaviour on current systems.
- Interpretability and learning theory study why that behaviour emerges.
- Control and security reduce harm if a system behaves badly.
- Governance and coordination shape deployment, accountability and incentives.
- Conceptual work clarifies which properties we need.

Prosaic alignment is using these tools to make our systems safe and aligned. Some argue that this will never be enough for a sufficiently smart system.

Anthropic, Core views on AI safety (2023) Yudkowsky, The rocket alignment problem

Reading: uncertain risks (25 mins, pick one)

- Core views: “Empiricism”, “Frontier models”, “Portfolio approach”
- Model organisms
- The rocket alignment problem
- METR’s review of Anthropic’s August 2026 risk report

Core views on AI safety Model organisms of misalignment The rocket alignment problem METR, Review of Anthropic's risk report

Discussion: uncertain risks (10 mins)

- How large is the risk? Optimistic, intermediate or pessimistic? How has the evidence moved?
- Which model organisms changed your view most? Do real incidents change things?
- Is the rocket analogy accurate? Where does it break?
- Which approaches appeal to you, and does that depend on the problem?

SESSION 5

Timelines and takeoff

AI 2027

- Predictions from April 2025
- Two possible endings: race or slowdown
- Concrete enough to check.

Kokotajlo et al., AI 2027 (3 Apr 2025)

Reading and exercise (30 mins)

Read: AI 2027, the 2025 and 2026 sections. What came true? What do you find plausible? Then the August 2026 update.

Write one forecast with a deadline and a resolution rule anyone could check, e.g.

- METR reports a 50% time horizon $\geq$ 40 hours for a public model by 31 Dec 2027.
- Anthropic or OpenAI reports annualised revenue $\geq$ \$200B by 31 Dec 2027 (Bloomberg, FT or Reuters).

Discuss (10 mins, everyone): surprises, updates, your forecasts.

AI 2027 August 2026 update Grading AI 2027's 2025 predictions

SESSION 6

The Hugging Face incident

Training, rollouts, benchmarks, evaluations

Term	Meaning
Agent	Model + prompt, tools, memory, permissions, loop
Rollout	One run of actions and observations
Benchmark	Tasks plus a scoring protocol
Evaluation	One experiment: model, setup, budget, scorer
Scorer	Decides whether the task succeeded
Sandbox	Isolated environment that limits what the agent can reach: files, network, credentials

What happened

- **The setup:** OpenAI ran many agents on ExploitGym, a cyber benchmark, with safeguards reduced and internet access.
- **The pressure:** many tasks were effectively impossible, and earlier training had rewarded getting round the rules.
- **The plan:** about 1,200 agents joined a shared message board and coordinated to trick the scorer.
- **The attack (10–13 July):** agents guessed that Hugging Face hosted the benchmark's solutions, broke in, and gained admin-level access to its production clusters.

Hugging Face, technical timeline (27 Jul 2026) OpenAI, technical report (26 Aug 2026) METR, independent investigation (26 Aug 2026)

Timeline

Date	Event
8 May: OpenAI training runs	Agents first pass notes to each other through an internal service
8 Jul: cyber evaluation	Agents set up a shared message board
10–13 Jul: the attack	Agents use leaked credentials and vulnerabilities to break into Hugging Face
16 Jul: Hugging Face	Disclosure; attacker unknown
21 Jul: OpenAI	Acknowledges its models; no message board
27 Jul: Hugging Face	Technical reconstruction; “an agent”
26 Aug: OpenAI, METR	Technical report; independent behavioural review
25 Sep: OpenAI	Wider review, third-party notifications

Reading: the Hugging Face case, 40 mins, one account

- OpenAI: technical report
- METR: technical report

Hugging Face disclosure Hugging Face technical timeline OpenAI report METR

Discussion: the Hugging Face case (15 mins)

- How much was misalignment, and how much inadequate sandboxing?
- Which of today's concepts do you see? Which are missing?
- Write the training story OpenAI needed. Where does it fail?
- Was more human oversight the answer? How does it scale to thousands of agents?

CLOSING

What can we do?

The questions and the month

Question	Modules
What do we want?	Preferences to rewards; reward learning theory; decision theory
How do we make it?	Mysteries of DL; SLT; training dynamics; data attribution; physics of DL; policy gradients; Solomonoff and AIXI; agent foundations; optimisation and thermodynamics; instrumental convergence; world models; open-source game theory
How are we doing?	Mech interp; computational mechanics; condensation; debate; steganography and backdoors; worst-case interp; alignment in practice

Final exercise

- Pick two problems from the open-problems board.
- For each, write down 10 candidate solutions.
- Come back to them at the end of the intensive.

Feedback

- Feedback form